HW4 for Modern Probability

Total Points: 100

1. [10 points] Let $X_1, X_2, \ldots$ be independent Bernoulli random variables with $\mathsf{P}(X_n = 1) = p_n \in (0, 1]$. Let
$$A_n = \sum_{k=1}^{n} p_k, \quad N_n = \sum_{k=1}^{n} X_k.$$
Assume $A_n \to \infty$. Show that
$$\sum_{n=1}^{\infty} \frac{\mathrm{Var}(X_n)}{A_n^2} < \infty,$$
and deduce that
$$\frac{N_n}{A_n} \to 1$$
almost surely.

2. Let X_n, $n \geq 1$, be arbitrary r.v.s on $(\Omega, \mathcal{F}, \mathsf{P})$ such that $\sum_{n=1}^{\infty} \varepsilon_n X_n$ converges P-a.s. for every deterministic sequence $(\varepsilon_n)_{n\geq 1} \in \{-1, 1\}^{\mathbb{N}}$. The goal is to show that $\sum_{n=1}^{\infty} X_n^2 < \infty$ almost surely.
   (a) [5 points] Let $(\xi_n)_{n\geq 1}$ be i.i.d. random variables on $(\Theta, \mathcal{G}, \mu)$ with $\mu(\xi_n = \pm 1) = \frac{1}{2}$. Let $(\tilde{\Omega}, \tilde{\mathcal{F}}, \tilde{\mathsf{P}}) = (\Omega \times \Theta, \mathcal{F} \otimes \mathcal{G}, \mathsf{P} \times \mu)$ be the product space. Using Fubini's theorem, show that
$$\tilde{\mathsf{P}}\Big(\Big\{(\omega, \theta) : \sum_{n=1}^{\infty} \xi_n(\theta) X_n(\omega) \text{ converges}\Big\}\Big) = 1,$$
   and hence for P-a.e. ω, $\sum_{n=1}^{\infty} \xi_n(\theta) X_n(\omega)$ converges for μ-a.e. θ.
   (b) [5 points] Use Kolmogorov's three-series theorem on $(\Theta, \mathcal{G}, \mu)$ to conclude that, for the ω obtained in part (a),
$$\sum_{n=1}^{\infty} |X_n(\omega)|^2 = \sum_{n=1}^{\infty} \mathrm{Var}_{\theta}\big(\xi_n X_n(\omega)\big) = \sum_{n=1}^{\infty} \int |\xi_n(\theta) X_n(\omega)|^2 \, \mu(d\theta) < \infty.$$

[Total for Question 2: 10 points]

3. [10 points] Let X be a random variable with a nonatomic distribution; that is, $\mathsf{P}(X = a) = 0$ for every $a \in \mathbb{R}$. Let
$$X_n = 2^{-n}[2^n X],$$
where $[x]$ denotes the largest integer $k \leq x$. Show that $X_n \Rightarrow X$ but $\|\mu_{X_n} - \mu_X\|_{TV} \not\to 0$.

4. [10 points] Let X_n and Y_n be defined on the same probability space. Suppose that $X_n \Rightarrow X$ and $Y_n \to c$ in probability, where c is a constant. Show that $X_n + Y_n \Rightarrow X + c$.

   Hint: You may either prove convergence of the distribution functions of $X_n + Y_n$ at every continuity point of the distribution function of $X + c$, or prove convergence of the characteristic functions.

5. Let $X_1, X_2, \ldots$ be i.i.d. random variables with an even continuous density f supported on $[-1, 1]$, with $f(0) > 0$.

   (a) [5 points] Recall that a random variable Z has a Cauchy distribution with scale parameter $c > 0$ if its density is

   $$g(x) = \frac{1}{\pi} \frac{c}{x^2 + c^2}.$$

   Compute the characteristic function of Z.

   Hint: Use the residue theorem.

   (b) [10 points] Show that

   $$\lim_{n\to\infty} n \int_{\mathbb{R}} \big[\cos \frac{\xi}{nx} - 1\big] f(x)\, dx = -\gamma|\xi|$$

   for some constant $\gamma > 0$. Identify the constant γ.

   (c) [5 points] Show that for every ξ,

   $$\mathsf{E} e^{i\xi \sum_{m=1}^n \frac{1}{nX_m}} = \exp\Big(n \log \int_{\mathbb{R}} \cos \frac{\xi}{nx} f(x)\, dx\Big) \to e^{-\gamma|\xi|}.$$

[Total for Question 5: 20 points]

6. Let X_1 and X_2 be two independent copies of a random variable X with $\mathsf{E}X^2 < \infty$. Suppose that

   $$\frac{X_1 + X_2}{\sqrt{2}} \stackrel{d}{=} X.$$

   (a) [5 points] Show that $\mathsf{E}X = 0$.

   (b) [5 points] Show that, for every $m \geq 1$, if $X_1, \ldots, X_{2^m}$ are independent copies of X, then

   $$X \stackrel{d}{=} 2^{-m/2} \sum_{k=1}^{2^m} X_k.$$

   (c) [10 points] Use the CLT to conclude that $X \sim \mathcal{N}(0, \sigma^2)$, where $\sigma^2 = \mathsf{E}X^2$. Include the degenerate case where $\sigma^2 = 0$.

[Total for Question 6: 20 points]

7. Let $X_1, X_2, \ldots$ be i.i.d. with $\mathsf{E}X_1 = 0$ and $\mathrm{Var}(X_1) = \sigma^2 \in (0, \infty)$. Let $S_k = X_1 + \cdots + X_k$ where $S_0 = 0$.

   Let N_n be nonnegative integer-valued random variables satisfying

   $$\frac{N_n}{n} \to 1 \quad \text{in probability.}$$

   Do not assume that N_n is independent of (X_k).

(a) [10 points] Fix $\delta \in (0,1)$. Use Kolmogorov's maximal inequality to show that

$$\mathsf{P}(|S_{N_n} - S_n| > \varepsilon\sqrt{n}) \leq \mathsf{P}(|N_n - n| > \delta n) + \frac{2\sigma^2\lceil \delta n \rceil}{\varepsilon^2 n}.$$

Deduce that $n^{-1/2}(S_{N_n} - S_n) \to 0$ in probability.

(b) [10 points] Show that

$$\frac{S_{N_n}}{\sigma\sqrt{n}} \Rightarrow \mathcal{N}(0,1).$$

Hint: Compare the characteristic functions of $S_{N_n}/(\sigma\sqrt{n})$ and $S_n/(\sigma\sqrt{n})$, and use part (a) and the CLT.

[Total for Question 7: 20 points]